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| **Fiscal Year:** | FY 2004 | **Task Last Updated:** | FY 12/09/2004 |
| **PI Name:** | Colwell, Joshua Ph.D. | | |
| **Project Title:** | Physics of Regolith Impacts in Microgravity Experiment (PRIME) | | |
| **Division Name:** | Physical Sciences | | |
| **Program/Discipline:** | FLUID PHYSICS | | |
| **Program/Discipline--Element/Subdiscipline:** | FLUID PHYSICS--Granular flows | | |
| **Joint Agency Name:** | | **TechPort:** | No |
| **Human Research Program Elements:** | None | | |
| **Human Research Program Risks:** | None | | |
| **Space Biology Element:** | None | | |
| **Space Biology Cross-Element Discipline:** | None | | |
| **Space Biology Special Category:** | None | | |
| **PI Email:** | josh.colwell@lasp.colorado.edu | **Fax:** | FY |
| **PI Organization Type:** | UNIVERSITY | **Phone:** | 303-492-6805 |
| **Organization Name:** | University of Colorado | | |
| **PI Address 1:** | Laboratory for Atmospheric and Space Physics | | |
| **PI Address 2:** | 392 UCB | | |
| **PI Web Page:** | | | |
| **City:** | Boulder | **State:** | CO |
| **Zip Code:** | 80309-0392 | **Congressional District:** | 2 |
| **Comments:** | | | |
| **Project Type:** | Ground | **Solicitation / Funding Source:** | 98-HEDS-03 |
| **Start Date:** | 03/01/2000 | **End Date:** | 11/30/2003 |
| **No. of Post Docs:** | 0 | **No. of PhD Degrees:** | |
| **No. of PhD Candidates:** | 0 | **No. of Master' Degrees:** | |
| **No. of Master's Candidates:** | 4 | **No. of Bachelor's Degrees:** | |
| **No. of Bachelor's Candidates:** | 2 | **Monitoring Center:** | NASA GRC |
| **Contact Monitor:** | | **Contact Phone:** | |
| **Contact Email:** | | | |
| **Flight Program:** | | | |
| **Flight Assignment:** | | | |
| **Key Personnel Changes/Previous PI:** | | | |
| **COI Name (Institution):** | Sture, Stein ( University of Colorado ) | | |
| **Grant/Contract No.:** | NAG3-2400 | | |
| **Performance Goal No.:** | | | |
| **Performance Goal Text:** | | | |
| **Task Description:** | The Physics of Regolith Impacts in Microgravity Experiment is a flexible and reusable experimental device for performing low velocity impact experiments in a vacuum and in the reduced gravity environment of NASAs KC-135 airplane. The experiment will incorporate design concepts from the space shuttle-based Collisions Into Dust Experiment (COLLIDE) but will be an entirely new apparatus optimized for maximum data return on the KC-135. The experimental apparatus is capable of performing 8 independent impact experiments per flight on the KC-135. Ground-based experiment are carried out in conjunction with the reduced-gravity experiments for comparison.<br>The purpose of PRIME is to study the physics of low velocity collisions and impacts into powders and dusty target layers simulating the surfaces of various planetary objects. A new experimental apparatus was developed to perform airplane-based reduced gravity low-energy impact experiments. This reusable experimental apparatus will be used with a variety of impact parameters to discover the relationships between ejecta properties, such as size and velocity | | |

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| | distributions, and impact parameters, such as impactor mass, impact velocity, and target regolith properties. These experiments will span the range of impact velocities from 10 m/s to 10 cm/s.<br><br>The results of the study will be used to better understand the physical and dynamical evolution of collisional disks in the solar system and the effects of low velocity disturbances to planetary regoliths. |
| **Rationale for HRP Directed Research:** | |
| **Research Impact/Earth Benefits:** | |
| **Task Progress:** | Task was completed in 2003. Final report was submitted. |
| **Bibliography Type:** | Description: (Last Updated: 12/05/2005) |